

Variational Protocol Dashboard

Product Requirements & Implementation Document (Phase 1)

Version: 1.0

1. Project Overview

Objective

Build a premium analytics dashboard for Variational Protocol that provides users with transparent visibility into:

- Protocol health
- Treasury holdings
- Open Interest
- Trading volume
- Market statistics
- Risk indicators
- Treasury activity

The dashboard should serve as a better user experience than raw blockchain explorers and provide insights that are not immediately visible through Arbiscan or the Variational website.

Phase 1 will be entirely frontend-focused and stateless.

No database will be used.

All data will be fetched directly from public APIs and blockchain data providers.

2. Product Goals

Primary Goals

Provide treasury transparency.

Provide protocol health metrics.

Provide market-level analytics.

Provide a professional investor-grade experience.

Create a foundation that supports historical analytics in Phase 2.

Non-Goals (Phase 1)

No authentication.

No user accounts.

No portfolio tracking.

No historical storage.

No database.

No custom alerting system.

No notification system.

3. Technical Architecture

Frontend

Next.js 15

TypeScript

TailwindCSS

shadcn/ui

TanStack Query

Recharts

Lucide Icons

Data Sources

Variational API

Provides:

- TVL
 - Open Interest
 - Volume
 - Market data
 - Funding rates
 - Loss refund metrics
-

Treasury Wallet

Wallet:

0x5E91B40467FB8902C46A7B6CB90482363188D645

Provides:

- Asset balances
 - Treasury allocation
 - Treasury activity
 - Token holdings
-

Price Feed

CoinGecko API

Provides:

- Token prices
 - Treasury valuation
-

4. Information Architecture

Navigation

Overview

Treasury

Markets

Analytics

5. Overview Page

Purpose:

Provide an executive summary of protocol health.

KPI Section

Display:

Treasury Value

TVL

Open Interest

24h Volume

Cumulative Volume

Markets Listed

Loss Refund Pool

Loss Refunds Paid (24h)

Market Snapshot Table

Columns:

Market

Mark Price

Funding Rate

Long OI

Short OI

Total OI

Top Markets Chart

Horizontal bar chart.

Metric:

Volume by market.

6. Treasury Page

Purpose:

Provide complete transparency into treasury assets and activity.

Treasury Summary

Cards:

Treasury Value

Total Assets

Largest Position

Wallet Age

Transaction Count

Asset Allocation

Pie Chart

Visual breakdown of holdings.

Holdings Table

Columns:

Token

Amount

Price

USD Value

Allocation %

Contract Address

Arbiscan Link

Treasury Activity

Recent transfers.

Columns:

Timestamp

Direction

Token

Amount

Counterparty

Transaction Hash

Direction Tags:

Incoming

Outgoing

7. Markets Page

Purpose:

Provide detailed market analytics.

Market Summary

Cards:

Highest OI

Highest Volume

Highest Funding

Lowest Funding

Markets Table

Columns:

Market

Price

Funding Rate

Long OI

Short OI

Total OI

Spread

Volume

Features:

Search

Sorting

Pagination

Sticky Header

Open Interest Visualization

Bar chart.

Display:

Market vs OI

Funding Visualization

Funding rates chart.

Positive values above baseline.

Negative values below baseline.

8. Analytics Page

Purpose:

Provide derived metrics and protocol health indicators.

Treasury Health

Metric:

Treasury / TVL

Formula:

Treasury Value ÷ TVL

Metric:

Treasury / Open Interest

Formula:

Treasury Value ÷ Open Interest

Metric:

Loss Pool Coverage

Formula:

Loss Refund Pool ÷ Open Interest

Market Concentration

Formula:

Largest Market OI ÷ Total OI

Risk Indicator:

Low

Moderate

High

Long/Short Imbalance

Formula:

Total Long OI ÷ Total Short OI

Status:

Bullish

Neutral

Bearish

9. API Layer

services/variational.ts

Responsibilities:

Fetch protocol metrics.

Normalize responses.

Provide typed data.

Functions:

getProtocolStats()

getMarkets()

getFundingRates()

getOpenInterest()

services/treasury.ts

Responsibilities:

Fetch wallet balances.

Calculate treasury value.

Retrieve treasury activity.

Functions:

getBalances()

getTransfers()

getTreasuryValue()

services/pricing.ts

Responsibilities:

Token pricing.

Functions:

getTokenPrice()

getMultiplePrices()

10. State Management

Use TanStack Query.

Caching Strategy:

Protocol Metrics: 60 seconds

Treasury Data: 60 seconds

Transfers: 30 seconds

Price Data: 60 seconds

11. Error Handling

Every page should support:

Loading State

Error State

Empty State

Automatic Retry

Manual Refresh

12. Responsiveness

Desktop First

Supported Widths:

1920px

1440px

1280px

1024px

768px

Mobile Friendly

13. Design System

Theme

Dark Mode Only

Colors

Background:

09090B

Card Background:

18181B

Border:

27272A

Primary Accent:

095BE5

Success:

22C55E

Danger:

EF4444

Warning:

F59E0B

Text Primary:

FFFFFF

Text Secondary:

A1A1AA

Typography

Font:

Inter

Weights:

400

500

600

700

Components

KPI Cards

Data Tables

Pie Charts

Bar Charts

Line Charts (future)

Badges

Tooltips

Dropdowns

Skeleton Loaders

14. Future Phase 2

Historical Snapshots

Database

Revenue Analytics

Treasury Growth

Volume Trends

Open Interest Trends

Funding History

Protocol Health Score

Alerting System

Treasury Change Tracking

Historical Asset Allocation

Investor Reporting

Success Criteria

The dashboard should answer the following questions within 10 seconds:

How large is the treasury?

What assets does the treasury hold?

How healthy is the protocol?

Which markets drive most activity?

What is the current open interest?

How concentrated is protocol risk?

How active is the treasury?

How much protection exists through the refund pool?

The dashboard should feel comparable to premium products such as DefiLlama, Token Terminal, Arkham, Hyperliquid Analytics, and Dune dashboards.